Biophysik Übung Woche 7

Synchrotron radiation for bio-materials

X-ray imaging, X-ray tomography, X-ray spectroscopy, X-ray scattering

Advantages of X-ray Radiation:

- → Short Wavelength: higher resolution compared to visible light, allowing detailed imaging of small structures like NP
- → Larger Penetration Depth: than electron or light microscopy.
- → Element-specific interaction depending on X-ray energy: Element mapping & 3D structures accessible
- → Non-Destructive Imaging: can visualize biological samples in near-physiological condition.

X-ray Tubes vs Synchrotron Radiation

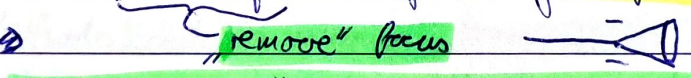
- → X-ray Tubes: Commonly used, low maintenance, and provide fixed wavelength with low intensity
- → Synchrotron Radiation: High intensity, high brilliance, Tunable wavelength, making it superior for high-resolution imaging and analysis. Requires specialized facilities for beamtime

Brilliance: Photons emitted per area, beam divergence, within bandwidth (BW)

$$\text{Brilliance} = \frac{\text{photons}}{\text{second} \cdot \text{mrad}^2 \cdot \text{mm}^2 \cdot 0{,}1\%\,\text{BW}}$$

Generation of Synchrotron Radiation

- → Electrons are accelerated to near the speed of light and guided in a circular path using bending magnets.



- → Radiation is emitted tangentially as the electrons are deflected by the magnets, with devices like wiggler & undulator used to increase brilliance and focus the radiation

→ thicker material

Why high brilliance?
↳ enhance resolution and focus
↳ improve signal to noise ratio

Synchrotron Light:
↳ the light that electrons emit that have been accelerated

Biophysikalische Nanostrukturen Woche 7

Nanoparticle - Based Drug Delivery

Benefits: (+) Therapeutic window will expand when approved drugs are applied as nanocrystalline formulation

(+) Large surface-to-volume ratio & loading with Drugs

(+) Combination of therapy and diagnostic: theragnostic vehicles

(+) Target delivery and controlled release: Size of materials same order as biological mechanisms, mitele cell-cell interactions

Challenge: Methods are required to monitor absorption, distribution, metabolism, excretion and toxicity of nanomaterials in biological settings

X-ray imaging techniques

→ High Spatial Resolution:

Synchrotron radiation can achieve imaging at subcellular levels, providing insights into the structure and organization of biological materials.

→ Temporal and Spectral Resolution:

Fast detectors allow for real-time imaging, and spectral resolution adds in elemental analysis, crucial for studying complex materials.

→ Depth Penetration:

Enables visualization of internal structures in their native state, which is important for studying biological specimens and inorganic NPs.

Methods such as scanning transmission X-ray microscopy (STXM) and STXM combined with tomography

⟶

Dual-Energy STXM: High Resolution, Elemental Specific Imaging, Quantitative analysis

Tomographic STXM: Dual-Energy STXM combined with Tomography

↳ can visualize: intracellular distribution & organelle segmentation

Bridging the Nanostructure World

Advanced Techniques

- **X-ray Absorption Near Edge Structure (XANES):**
  - → Principle: ~~X-ray [illegible]~~ When X-rays are absorbed by a material, they generate an excited photoelectron and create a hole
  - → Process: The hole is filled by an electron from a higher energy level, emitting a fluorescent photon.
  - → Absorption Edge: This is the specific energy level where significant absorption occurs (Energy level where spectral absorption occurs)
  - → Fine Structure: The emitted photon is scattered by neighboring atoms creating a fine structure near the absorption edge, which is used to study material properties
  - → Data Analysis: By comparing the measured data with known standards, the characteristics of the material can be determined.

- **X-ray Fluorescence Spectroscopy:**
  - → Principle: X-ray eject an electron from an atom, creating a hole that is then filled by another electron, emitting a fluorescent photon.
  - → Element Specific: The emitted fluorescence is unique to each element, allowing for elemental analysis
  - → Application: 3D elemental maps, Cancer Research

- **Comparison**

  XANES: for analyze the fine structure of the material near the absorption edge, providing detailed info. about the electronic and structural environment of specific elements.

  X-ray Fluorescence Spectroscopy: for elemental analysis and mapping, identifying the presence and distribution of elements within a sample.

Biofabricated Vascular Grafts 2024-7

Radiation Damage

→ higher dose = higher resolution

→ High doses of X-rays can damage biological samples, necessitating the use of protective measures like cryogenic preparation and radioprotective agents to minimize harm during imaging

→ Damage reduces resolution

Natural, biological materials

Biomolecules (proteins, DNA) :
- → defined 3D structure
- → atomically precise
- → C, H, N, O, S, etc.
- → can be repaired / regenerated
- → soft (elastic / stretchable)

Inorganic Component :
- → eg: calciumphosphate, calciumcarbonat
- → mineralized within the organic matrix
- → hard (stiff, rigid)

⇒ Mostly structural materials, mechanical properties are important

Hierarchically organized: Matrix with inorganic component

Example Bone Material

Structural Overview: Collagen Molecule > ... > compact bone / spongy bone

* Procollagen: three intertwisted protein chains ⇒ // more details 7.17 - 7.20

* Tropocollagen: removal of loose ends yields tropocollagen

Micro X-ray computer tomography ~~and tensor~~ and tensor tomography provide 3D reconstructions and detailed structural analysis of bone and other biomaterials

micro cracks in bone

↳ Tensor Tomography: SAXS & WAXS

↳ Both: ↳ information about larger nanostructures and their organization | ↳ insights into the atomic and molecular structure of bone

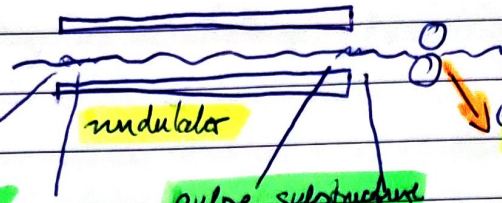
In vivo X-ray cone-tomography

→ Ultrafast X-ray microtomography combined with motion analysis

X-ray Free Electron Laser (XFEL)

→ High brilliance, short pulses

→ XFELs produce extremely short pulses of X-rays, allowing for time resolved studies and high-resolution imaging of individual particles and molecules. → pivotal for capturing dynamic processes and detailed structure info

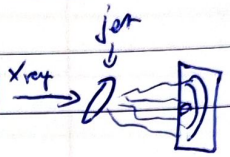


highly compressed electron pulse emits spontaneously

pulse substructure results in coherent superposition

! e⁻ in, X-rays out !

Serial Femtosecond Crystallography (SFX)



Process:

Involves injecting nanocrystals through a liquid jet and hitting them with X-ray pulses to collect diffraction patterns. Enables the determination of high-resolution structures of biomolecules.

Coherent X-ray Imaging (CXI)

High Resolution:

Allows for single-particle imaging with resolutions down to 3 nm, providing unprecedented detail in study of NPs and biological samples.

→

Biophysical Chemistry II

Conclusion

- Synchrotrons: generation of intense X-ray radiation
- X-ray based techniques have several advantages such as spectral resolution, penetration depth, time-resolved measurements
- Radiation damage is a challenge, especially when doing in vivo data collection
- Some techniques utilize absorption edges of elements present in the sample
- X-ray free electron lasers enable new types of experiments such as single particle imaging

Bridget Wanshura WS#8

Electron Microscopy

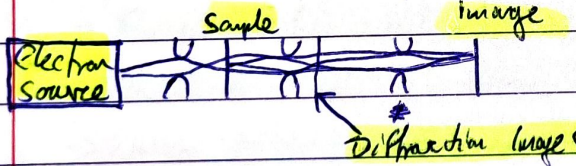
Motivation:
- -> High resolution imaging of nanoscale objects possible, down to <1nm
- -> Combines structural information with elemental composition analysis
- -> Strong interactions with samples: thin samples are imaged
- -> Requires: -> Maintenance of all parts (eg: cooling)
  - -> High vacuum for samples
  - -> Stable construction site

Transmission Electron Microscopy (TEM)

TEM: Transmission method for structure imaging of thin samples at (atomic) extremely high resolution.
It uses an electron beam to pass through thin samples, providing details on interatomic distances and crystal structure.

Determination of: interatomic distances, atomic crystal structures, morphology, defects

Resolution: limited through focus of electromagnetic lenses, scattering of electrons with residual gas



Two Modes: -> captures detailed images of sample morphology and structure

1) Imaging:
- Bright field: Scattering and diffraction contrast, regions with high absorption and scattering will appear dark
- High-angle annular dark field (HAADF): Unscattered beam is removed, scattered beam is imaged. Strongly scattering atoms will appear bright

2) Diffraction: selected area electron diffraction (SAED) gives info about crystallinity of the sample and possibly lattice structure
   - -> provides structure
   - -> analyzes the atomic arrangement and crystallography of the sample

Biophysik Demonstration Wdh 8

TEM analytics

- EDX & EELS

EDX: e⁻ source → sample → CCD

Spectrum: qualitative & quantitative Info

Image: Structure Information

EDX = Energy Dispersive X-Ray

used for: Elemental Composition

Principle:

→ Sample is hit with high-energy primary e's

→ Inner core e's are removed

→ Relaxation of higher shell e's into holes

→ Emission of characteristic, element-specific X-ray radiation

→ Detection with Si(Li) scintillator crystal and photomultiplier

⇒ Analytics of heavy atoms

EELS:

EDX: e⁻-source → sample → CCD

Image: quantitative Info with spatial resolution

EELS = Electron Energy Loss Spectroscopy

Principle:

used: electronic structure & chemical composition

→ Sample is hit with high-energy primary e's

→ Inelastic scattering (losing energy to sample)

→ Energy filter and detector to determine energy loss

→ Information about electronic state and next neighboring atoms

→ Energy-filtered imaging to visualize element distribution

⇒ Analytics of light atoms

What are the differences between X-rays and electron radiation?

X-ray: electromagnetic waves with high penetration power

used extensively in imaging & diagnostic applications

Electron Radiation: high-speed particles with limited penetration, used in surface-level treatments and material processing

↳ higher resolution

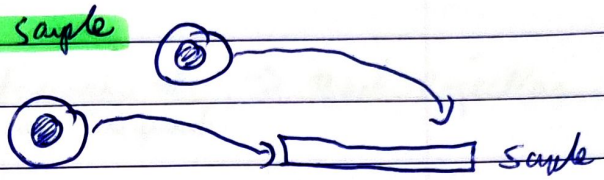
Biological Nanoobjects WH8

Staining in electron microscopy

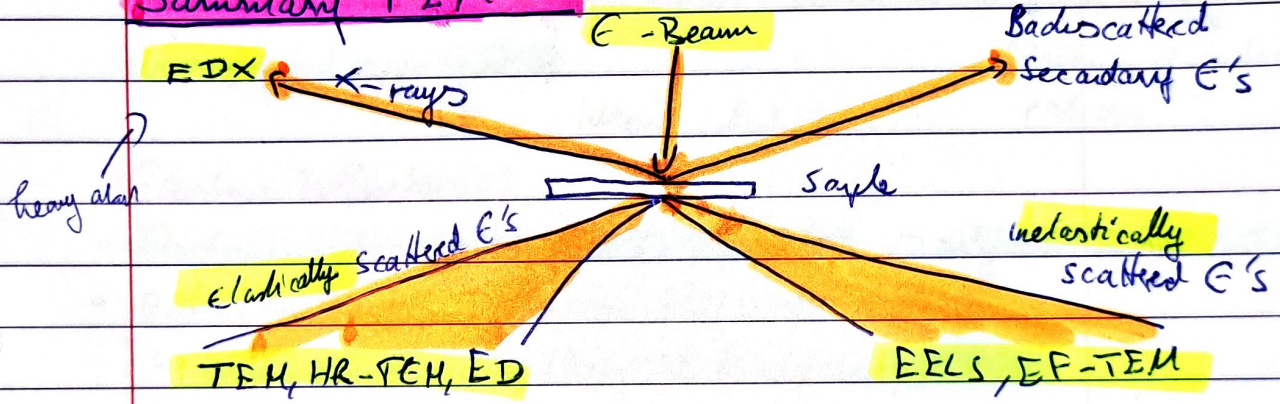
- Electrons are scattered ⇒ Heavy atoms provide more contrast
- Biological structures do not contain many heavy atoms ↯
  ↳ stain with eg. uranyl
  → radiation damage
- Negative Staining:
  Dark: Heavy Atoms, Bright: Protein

Cryo EM better

How to stain: Add a drop of stain! ↗ eg: Gold!
↳ use protein containers for encapsulated NPs
↳ does those can then target a specific biological region!
(+) Contrast at target sample



Summary TEM



Staining: Improve contrast for biomolecules

↓
light atoms

Biophysical Characterization WS #8

Cryogenic Electron Microscopy (Cryo EM) pipeline

- Pipeline:

Biochemical Preparation → Cryo EM sample Preparation → Imaging → Data Collection

↓

Model ← Structural fitting ← Reconstruction ← Image Processing

→ Can image highly sensitive biological samples: low electron dose, low signal to noise ratio

- Data Processing:

Projection → Particle Finding → Averaging and Class assignment → Back Projection

Goal: Electron density, then fit a model into density (cf. protein crystallography)

Cryo EM is not a diffraction method, combination of local imaging with averaging and reconstruction

↳ averaging of individual particles

ideal method

Electron Diffraction

- Electrons interact strongly with crystals ⇒ only thin crystals: TEM
- Rotation Electron Diffraction (RED) for Zeolites
  - ↳ rotate sample and collect diffraction patterns
  - ↳ Application in HR-TEM (SAED)

Serial Protein Electron Diffraction

- Protein crystals deposited on grid (Holds crystal in place)
- Nanocrystals vitrified (rapidly frozen)
- Mapping of crystal positions // scanning to identify random crystal locations
- Data collection: Diffraction // selectively addressing mapped crystal positions
- No rotation required

Right frame rate

↑ = diffraction data stack

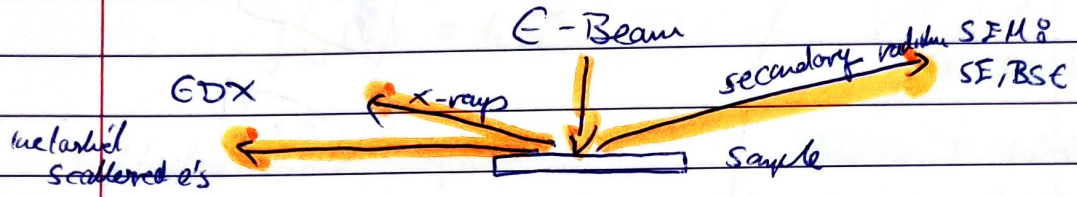
a) Scanning to identify random crystal locations → b) selectively addressing mapped crystal positions

Bibliothek Unterlagen WdH 8

Scanning Electron Microscopy (SEM)

used for analyze surface structures and composition of materials

- Sample is hit with low-energy e's
- E-Beam with dimensions of 1-10 nm scans the surface
- Secondary E's (SE): emitted from the surface of the specimen and provide detailed topographical information
- Backscattered E's (BSE): reflected back from the sample. provide compositional contrast because heavier elements scatter more e's back



↳ also mode in TEM: STEM

Summary of characterization methods

- Nanoparticles with ligand shell:
  - → TEM for inner core → stained TEM for ligand shell
  - → DLS / Zeta for hydrodynamic diameter → SAXS for core size
  - → HRTEM / STEM for crystal structure
- Assemblies of nanoparticles:
  - → SAXS for distances between NP's in solution → Small angle X-Ray Diffraction for superlattices
  - → SEM for morphology
- Protein - NP composites:
  - → stained & unstained TEM → DLS / Zeta
  - → Protein design / protein crystallography
- Interaction with biological environment:
  - → Synchrotron radiation
  - → Electron microscopy

Biomineralization Week 9

Biomineralization → eg: shells, pearls, bones

Biomineralization:
A process by which living organisms produce minerals to harden or stiffen existing tissues.

eg: bone, teeth, shells
- bone → collagen fibers & hydroxyapatite crystals
- shells → calcium carbonate

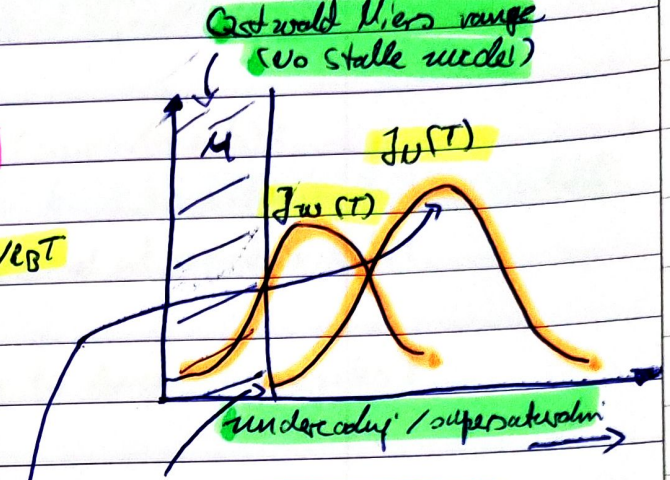
Nucleation rate and growth rate

Nucleation rate: → on a bulk crystal

$$J_N(T, \Delta G) = J_0 e^{-(\Delta G_c + \Delta G_D)/k_B T}$$

Growth rate: → growing crystal

$$J_W(S) = k S^x$$



- Nucleation rate depends on temperature & supersaturation

formation of stable nuclei from here growth due to coalescence or Ostwald ripening

- Crystal growth rate depends on:
(pure solutions without influence of additional ions or additional solid phases)

→ Diffusion control:
Diffusion from bulk solution to the crystal surface is the rate-determining step

→ Kinetic control: (activation energy)
Growth rate depends on the E_A of chemical processes, which take place before ions are incorporated into the lattice.
eg: biomineralization

Biohybrid Nanostructures WS 19

Mechanisms of Biomineralization

1) Control over the crystal nucleation:
Organisms can control the initial formation of mineral crystals by providing nucleation sites.
2) Control over crystal growth:
After nucleation, the growth of crystals is regulated by the organism, ensuring proper size and shape.
3) Crystallization in confined spaces (compartments):
Confined spaces, such as cells or specialized compartments, to ensure control
4) Formation of composite of inorganic mineral and organic matrix:
create strong durable materials
5) Assembly of single biocrystallites to larger aggregates:
○ + ○ → ◯

(=)

Construction of a city:
Planning: Formation of supramolecular areas that are intended for later crystallization
Foundation: nucleation also a defined location in a defined orientation
Building: crystal growth in defined compartments
City: assembly individual crystals

Y : Biotinylated Katas
⊛ : Katas
○ : Streps

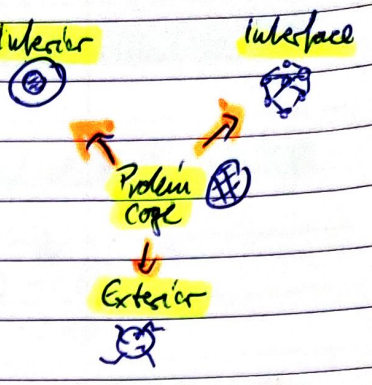
• Formation of compartments through pre-organization:
1) Self-organization of protein molecules:
↳ creates structural frameworks that provide a physical extent for mineral deposition
2) Regioselective nucleation:
→ matrix induced: specific activation of nucleation, but crystal structure and orientation is determined by chemical factors
→ matrix mediated: matrix determines crystal structure and orientation
↳ initiates & controls the location and characteristics of mineral crystal formation.
↳ nucleation at specific, controlled location on a biological template
1) collagen matrix in bone
2) precise mineral deposition in bones relies specific sites on the matrix template provide nucleation

→ biomineralization of calcite + fluorite in mollusk shell

Biological Nanostructures Wk #9

Protein Containers

- Protein Containers:
  - spherical protein assembly
  - consist of similar building blocks
  - hollow sphere
  - highly symmetric



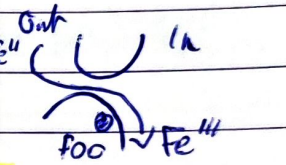
- For artificial materials:
  3 ways to modify protein containers

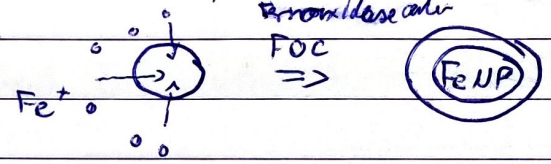
- viral: container for viral genome, encapsulates the virus RNA
- non-viral: ferritin: iron storage, ferrochelatase synthase: enzyme

→ can act as containers for mineral formation, providing a controlled environment for biomineralization.

→ template for synthesizing nanoparticles

eg: ferritin → purpose: iron storage protein





inner surface:
negatively charged due to AA Glu & Asp

→ electrostatic: gradient may funnel the positively charged ions into the cavity

↳ two subunits:
heavy chain (H) with ferroxidase center and light chain (L) with nucleation center.

H: Catalyzes the rapid oxidation of Fe^{2+} to Fe^{3+} (ferroxidase), preventing iron toxicity and preparing it for storage

L: Facilitates the slower and controlled aggregation of Fe^{3+} into a stable mineral core, ensuring safe long-term storage of iron.

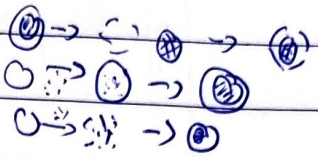
24 monomers 432 symmetry
↳ octahedral

Why Fe in Body?
↳ prevention of iron deficiency => needed for Hemoglobin

Biohybrid Nanostructures WdH 9

Why use a protein container for NP synthesis?

- Size - constrained synthesis of NPs: the container provides an upper limit for NP dimensions.
- Mild conditions for NP synthesis
- Final composite has an atomically defined outer shell (in contrast to NP ligand systems)
- Biocompatible shell, container can be further functionalized with targeting ligands, makes cargo water soluble



Magnetic iron oxide, oxide ferritin

→ cancer recognition peptide on the outside, targeting melanoma cells

more eg's in ferritin: silver, PbS quantum dots

Caveolin

Biomineralization: molecules with desired crystal structure, morphology and orientation

Biomolecule can control the inorganic component

Protein containers are size-constrained reaction vessels for NP-synthesis

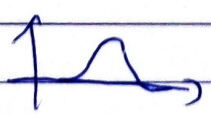
Protein redesign enables introduction of new features

Perform your tools at the histogram for the size distribution:
Is this the expected results from growth inside a confined cavity?

→ yes if „steel" not too wide range

or → not mature → Gaussian

→ exceeds possible limit of 6 nm ☺



Biomolecule Nanostructures Week # 10

Nanomaterial synthesis inside cells and collective interactions in nanomaterials.

-> Nanomaterial synthesis inside bacterial cells
-> Biogenic synthesis of nanocrystals (inside/outside cells, with bacteriophages)
-> Challenges

Synthesis of NPs

Properties depend on: size, shape and crystallinity

-> Control these parameters during synthesis
=> Focus: Semiconductor nanocrystals

Motivation to use microorganisms for nanomaterial synthesis

Chemical Routes	Biogenic synthesis with microorganisms
(-) energy intensive (high temp's)	(+) Mild reaction conditions, simple precursors
(-) Hazardous and toxic chemicals as capping agents or solvents	(+) Green chemistry & no solvents needed
(-) multi-step route	(+) Bacteria easy to produce, maintain & multiply
(-) yield and high cost	(+) Genetically tune nanomaterial properties, inducing size and shape

Yeast cell (Eukaryotic) vs Bacteria cell (Prokaryotic)

- Eukaryotic cells more complicated than prokaryotic cells
- Insertion of plasmids is possible, contains construction plan of heterologous protein for production
- E. coli for protein production

Biological Nanostructures Week 10

Synthesis of CdS nanocrystals in yeast cells

- Yeast cells synthesize phytochelatins.
  - Cysteine-rich peptides that bind heavy metal ions
- Nanocrystals formed both inside and outside the cells
- Mechanism:
  - → pre-complexation: Cadmium ions (Cd^{2+}) initially form complexes with phytochelatins
  - → transport and incorporation: transported to the vacuole, where S^{2-} (sulfide) are incorporated to form CdS nanocrystals
- Challenge:
  - → Multi-compartment mechanism: the synthesis mechanism in yeast involves multiple cellular compartments, making it complex and difficult to tune

Synthesis of CdS nanocrystals in E. coli

- Synthesis of CdS nanocrystals by incubation of E. coli with $CdCl_2$ and Na_2S solution.
- Depends on the growth phase of the E. coli cells, which affects the efficiency and size of the nanocrystals?

⇒ No clear correlation between amount of free thiols (during cell growth / strain type) and nanocrystal formation

→ Challenge:

Understanding the role of biological variables, such as free thiols and strain differences, in influencing the synthesis process

Bielefeld University Week 10

Engineering of E. coli with yeast ligands for CdS synthesis

- Expression of yeast phytochelatin synthase in E. coli
- second enzyme (cor-transformation): synthesis of glutathione, the precursor of PCs
- Nanocrystal synthesis:
  - ↳ Incubation with $CdCl_2$ during exponential growth

Recombinant E. coli as biofactory for nanomaterials

- E. coli co-expressing PCs and metallothioneins (small cysteine rich-protein)
  - ↳ can both bind heavy metals & facilitate nanomaterial synthesis
- 34 elements seperately tested for synthesis
- Nanomaterials (NMs) are crystalline or amorphous, formed inside/outside
- also two-element / three-element nanomaterials tested
  - ↳ Thermodynamic stability predictions were employed to predict viable combinations of elements for nanomaterial synthesis.
- Size of nanomaterials: concentration of precursor
- ligands: proteins & peptides, aliphatic ligands MTs & PCs

Synthetic gene circuit for doping ZnS nanocrystals with Mn^{II}

Shewanella: an aerobic marine bacteria with dissimilatory metal reduction (metal as electron acceptor)

↳ engineered strain: expression of cytochrome complex MtrCAB can be controlled

→ Reduces $Mn^{III/IV}$ to Mn^{II}, doping of ZnS nanocrystals

TMAO: → controls the expression of the cytochrome complex

↳ concentration controls the amount of Mn^{II}

→ ZnS nanocrystals outside the shell

Biological Nanostructures Week 10

Synthesis of arsenic sulfide nanotubes

Shewanella

-> Starting materials: Arsenate and Thiosulfate

-> Bacteria reduce As(V) to As(III) and $S_2O_3^{2-}$ to S^{2-}

-> Nanotubes are formed outside cells, templated by polysaccharide

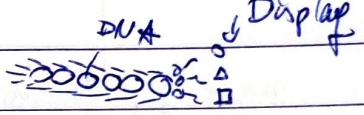
↳ luminescence and metallic conductive behavior

Synthesis with cell filtrate: CdS at high NaCl concentration

Halobacillus: extremophile halophile bacteria strain, tolerates high NaCl concentrations

- Synthesis of CdS with cysteine as precursor in presence of high NaCl concentrations
- Nanocrystals are synthesized in supernatant, easier to separate from cells

=> CdS nanocrystals tolerate high salt concentrations



M13 bacteriophage and phage display

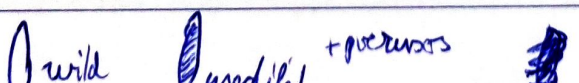
- M13 is a filamentous bacteriophage that infects E. coli bacteria. It is a virus composed of single-stranded DNA etc.
- M13 can be easily manipulated (genome) to display peptides or proteins on its surface.

↳ Phage Display is a molecular technique where foreign peptides or proteins are expressed on the surface of bacteriophages

- M13 can be used as templates for the synthesis and assembly of nanomaterials.
- By displaying specific peptides on their surface, phages can bind to and organize inorganic materials into desired structures

↳ eg: ZnS Nanocrystals Wires [↳ eg: tuner peptides -> Li Battery (cathode)]

Phages were genetically modified to display peptides that have a high affinity for ZnS nanocrystals.



Biological Nanomaterials Week 10

Library of mineralization peptides

Various peptides for different materials

⇒ control of crystal structure (phase) possible

→ can lead to nanocomposites

Challenges for inorganic material synthesis

-> Most studies focus on one material or one system -> systematic knowledge gap

-> Optical characterizations of nanocrystals issues

-> Ligand shell for nanocrystals is not well-defined and difficult to analyze

-> Only water-stable nanomaterials possible

-> Cell is complex environment, difficult to control, traces of elements can be present

Microorganisms for patterning of materials

Engineering E. coli for -> hierarchical assembly of inorganic materials

↳ E. coli produces curli protein

↳ Curli has peptide tags for recognition of inorganic materials

↳ Assembly of AuNP and CdTe nanocrystals

↳ in complex and ordered structures

Applications:

Biomining: • Recycling of electronic waste

↳ Microorganism reduce dissolved gold ions to gold

• In Microgravity

↳ Microorganism mine rare earth elements from rocks on ISS

Conclusion

- Synthesis of semiconductor materials feasible with bacteria cells
- Mostly thermodynamically stable products formed
- Mineralization peptides and templates could offer access to desired nanomaterial morphology (shape, size) and crystal structure

Bridging Nanofabrication Week 10

Nanoparticle Superlattices

Nanoparticle Superlattices:
Ordered arrays of NP arranged in a periodic structure similar to atomic crystals but at nanoscale.

Requirements for Superlattice formation:
-> short distances
-> precise arrangement
-> structural discretion

NP as building blocks:
-> Organic ligands
-> DNA ligands

Future: Optoelectronics, Spectral conversion, Sensors, Catalysis

Plasmonic Supercrystals

Plasmonic coupling: occurs when the electromagnetic fields of neighboring gold NPs interact, leading to collective oscillations of their surface electrons (Hot spots)

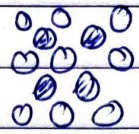
SERS: Raman Spectroscopy with analytes (used for that)
(electromagnetic field is enhanced)

-> Localized surface plasmons enhance the signal up to 10^{10} hot spots even more
-> Detection of biomolecules at single molecule level

=> Plasmonic supercrystals leverage the collective interactions of gold NP to create intense electromagnetic field known as hot spots. Those hot spots are crucial for SERS dramatically increasing the sensitivity of Raman signals and enabling the detection of biomolecules at single molecule levels.

Biological Nanotechnology Week 10

Magnetic Superlattices

Assembly of Fe_3O_4 and Co =>  -> magnetic

Conclusion

-> Arrangement of NP's into superlattices can produce novel properties

-> Magnetic properties can be altered by assembly into superlattices

-> Good & High long-range order and large domain sizes

-> Plasmonic coupling produces hotspots